

Smelting is one of the most common ways of extracting metals out of batteries. However, in case of lithium, traditional smelting leads to its extraction in the form of a mixed byproduct. Extracting lithium from the byproduct is complicated and a costly affair. Traditional smelting, hence, leads to procurement of barely 30 per cent of the total lithium available.

Given the expense associated with the process and the infrastructure required for actual procurement, investments have not yet proved economical. Car makers as well as battery research firms across the globe are attempting to bring modifications to this process, and some newer firms are starting deployment of such services.

Various governments are also framing regulations to push EV battery recycling. For instance, the EU has put in place a regulation that mandates battery makers to finance and initiate collection, treatment and recycling of electric batteries. This has kickstarted a number of partnerships between battery producers and car manufacturers. One example is Belgian battery producer Umicore's collaboration with Toyota and Tesla. The UK government has also banned incineration or landfill disposal of all types of industrial and automotive batteries.

China is also cautious about its EV production. It is all set to put forth a strict set of regulations for battery reuse. China's Ministry of Industry and Information Technology had noted earlier this year that, 17 cities and regions, starting with Beijing, will encourage automakers to partner with battery makers, scrap merchants and used car dealers to build recycling outlets in all locations. The



Apart from battery recycling, India needs to improve its roads and charging infrastructure to accommodate EVs

ministry has strict reservations about new battery producers.

The regulation also plans to provide incentives and benefits to encourage existing battery dealers to kickstart the recycle process. To ensure that all used EV batteries are accounted for, it has prepared a draft for rules to develop a traceability management platform, for the entire lifecycle of batteries. With these regulations, China plans to ensure that its target of two million EV outputs by 2020 do not backfire as a bigger environmental problem on the side.

India's progress

India started domestic production of lithium batteries this year, in Tamil Nadu through Raasi Solar. The technology has been developed by CSIR - Central Electro Chemical Research Institute (CECRI). This marks pivotal progress towards procuring indigenous batteries at a lower cost with better control rate. However, strategies for recycling of batteries need to be chalked out simultaneously.

"There is an elephant in the room. No one is talking about recycling lithium batteries. It is critical that this subject is figured out quickly, else we will end up removing one problem and getting into another bigger one," says Biju Bruno, managing director, Greenvision Technologies.

The optimistic news is that discus-

sions have started, and some organisations are taking steps to look into this challenge. For instance, Mahindra Electric has expressed that it plans to enable EV battery recycling, in a method similar to the recycling of cellphone batteries, with the help of a supply partner.

C. Narasimhan, chairman, Raasi Solar, during the launch of the first battery production plant, announced his plans to set up a 300MW plant, which will focus on lithium battery recycling, in addition to battery assembling and cell manufacturing. Research firms like CSIR-CECRI and others are also looking for the best techniques and infrastructure for the same.

White papers have been submitted suggesting mandating minimum requirements for recycling materials and incentivising recycling investments. Plans are yet to mature, and a long road lies ahead to be covered for India.

For popularising EVs in India, many other challenges need to be addressed, including upgrade of road conditions, improvement of charging infrastructure, cost control of vehicles and more. Big players like Hyundai Motors, Maruti Suzuki and Toyota are set to bring out their range of commercial EVs from next year onwards. It is important for India to be prepared with all facilities to deliver the best EV experience to its citizens. **EFY**

Lithium batteries often pile up in landfills, which should be avoided, especially in case of EV batteries

